Chapter 2: The Application Layer

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Exercise 1

Suppose within your Web browser you click on a link to obtain a Web page. The IP address for the associated URL is not cached in your local host, so a DNS lookup is necessary to obtain the IP address. Suppose that n DNS servers are visited before your host receives the IP address from DNS; the successive visits incur an RTT of $RTT_1, ..., RTT_n$.

Further suppose that the Web page associated with the link contains exactly one object, consisting of a small amount of HTML text. Let RTT_0 denote the RTT between the local host and the server containing the object. Assuming zero transmission time of the object, how much time elapses from when the client clicks on the link until the client receives the object?

The Total amount time to get IP address is

$$RTT_1 + RTT_2 + \dots RTT_n$$

onece the IP address is known, RTT_0 elapse to set up the TCP conneetion and another RTT_0 elapses to request and receive the small object. The ~~total~~ respond time is

$$2RTT_0 + RTT_1 + RTT_2 + \dots + RTT_n$$

Exercise 2

Consider distributing a file of $F = 20$ Gbits to N peers. The server has an upload rate of $\mu_s = 30$ Mbps, and each peer has a download rate of $d_i = 2$ Mbps and an upload rate of μ. For $N = 10$, 100, and 1,000 and $\mu = 300$ Kbps, 700 Kbps, and 2 Mbps, prepare a chart giving the minimum distribution time for each of the combinations of N and μ for both client- server distribution and P2P distribution.

1. client server Distrabution time : $T_{cs} = \frac{F}{\mu_s} * N$
2. peer to peer distribution time: $T_{p2p} = max\left(\frac{F}{\mu_s}, \frac{F}{N \times \mu}\right)$

Given: $F = 20 \times 10^3$ Mbits, $\mu_s = 30$ Mbps, $d_i = 2$ Mbps

μ: 0.3 Mbps, 0.7 Mbps, and 2 Mbps, N: 10, 100, 1000

N	μ (Mbps)	Client – server Time (minutes)	Peer to peer Time (minute)
10	0.3	111.11	111.11
10	0.7	111.11	47.62
10	2	111.11	16.67
100	0.3	1111.11	11.11
100	0.7	1111.11	11.11
100	2	1111.11	11.11
1000	0.3	11111.11	11.11
1000	0.7	11111.11	11.11
1000	2	11111.11	11.11

Exercise 3

Consider a DASH system for which there are N video versions (at N different rates and qualities) and N audio versions (at N different rates and qualities). Suppose we want to allow the player to choose at any time any of the N video versions and any of the N audio versions.

a) If we create files so that the audio is mixed in with the video, so server sends only one media stream at given time, how many files will the server need to store (each a different URL)?

If each file contain a mix of one audio version and one video version, the server needs to create a file for every possible combination of audio and video versions. since there are n version of video version. Since there are N versions of video and N version of audio, and each combination of a vedio and an audio version needs its own unique file. the total number of files is given by: Numbers of files $= N \times N = N^2$
Therefore, the server must store N^2 unique files, each accessible through a different URL.

b) If the server instead sends the audio and video streams separately and has the client synchronize the streams, how many files will the server need to store?

N Video files: One for each version of the video

N Audio files: one for each version of the audio

The server needs to store files for all video versions plus all audio versions:

Number of video files $= N$ number of audio files $= N$ Total number of files $= N + N = 2N$

Thus, the server needs to store $2N$ files.